

Redirection of SARS-CoV-2 to phagocytes by intranasal sACE2-Fc as a universal decoy confers complete prophylactic protection

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This manuscript presents a **valuable** antiviral approach using an engineered ACE2-Fc fusion protein that demonstrates broad-spectrum neutralization capacity against SARS-CoV-2 variants and achieves significant prophylactic protection in animal models through a novel Fc-mediated phagocytosis mechanism. The study provides **convincing** evidence for protective efficacy through rigorous in vivo validation in mice, mechanistic characterization via biodistribution studies and macrophage depletion assays, and demonstration of antibody-dependent cellular phagocytosis as the primary clearance mechanism. However, there are some gaps that require attention, including the need for comparison with a previously reported ACE2 decamer, inclusion of control molecules, insufficient discussion of potential limitations such as off-target binding and immunogenicity risks, and lack of clarity regarding certain methodological aspects.

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Abstract

The rapid evolution of SARS-CoV-2 and other respiratory RNA viruses limits the success of current vaccines and antibody-based therapies. Engineered decoy receptors based on soluble angiotensin-converting enzyme 2 (sACE2) offer promising alternatives. Clinical-grade recombinant sACE2 inhibits SARS-CoV-2 replication *in vitro* but shows limited clinical success.

This study reports an optimized sACE2 mutant fused to human IgG1 Fc (B5-D3), which redirects virus–decoy complexes to lysosomal degradation in macrophages. Intranasal prophylactic delivery of B5-D3 confers complete protection in SARS-CoV-2-infected K18-hACE2 mice. Abrogation of Fc effector functions compromises antiviral protection, indicating that Fc-mediated uptake of virus–decoy complexes is critical. Transcriptomic analysis suggests that B5-D3 induces early immune activation in lungs of infected mice. Bio-distribution and flow cytometry reveal selective targeting of airway phagocytes. *In vitro* assays confirm lysosomal degradation of virus–decoy complexes by macrophages without productive infection. These findings reveal a distinct antiviral mechanism via phagocytic clearance, supporting refined regimens for decoy treatments against SARS-CoV-2 and potentially other respiratory viruses.

Introduction

The incessant evolution of severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) and frequent breakthrough infections during the coronavirus disease 2019 (COVID-19) pandemic underscore the critical need for effective antiviral strategies that are less susceptible to immune escape than conventional vaccines and monoclonal antibody (mAb) therapies [44].

Soluble angiotensin-converting enzyme 2 (sACE2) therapies, which employ recombinant forms of the human angiotensin-converting enzyme 2 (ACE2) receptor—the primary binding site for the SARS-CoV-2 spike protein [15,24,32,34,44]—as viral decoys, have emerged as a promising alternative [31]. However, an early clinical version (amino acid [aa] 1–740, APN01) showed limited therapeutic benefit [51] and raised safety concerns about interference with endogenous renin-angiotensin system (RAS) [36]. Subsequent protein engineering greatly improved the pharmacological properties of sACE2, including fusion with a human IgG1 Fc domain (sACE2-Fc) to enhance serum half-life [25], and mutagenesis to enhance spike-binding affinity [4,11,33] and abolish enzymatic activity [11,26,33]. Potent sACE2-Fc mutants have shown broad-spectrum neutralization against SARS-CoV-2 variants in animal models [7,14,18]. However, the efficacy was limited. Despite the evidence suggesting a role for Fc-mediated effector functions [7], underlying immune mechanisms remained poorly understood. Further investigation is pivotal to advance the development of decoy-based antivirals and harness their full potential.

In this study, we engineered and characterized a potent sACE2-Fc mutant (B5-D3) with just two mutations that enhance spike-binding and eliminate enzymatic activity. Broad-spectrum neutralization capacity was confirmed by *in vitro* neutralization assays. Markedly, stepwise examinations of various administration routes and time points identified intranasal (IN) prophylaxis as the most effective regimen for B5-D3, which conferred complete protection against SARS-CoV-2 infection in K18-hACE2 mice across age groups. To understand how sACE2-Fc decoys influence viral fate, we carried out systematic, mechanistic investigations through transcriptomics, bio-distribution, and phagocytosis analysis. Our results revealed that IN-delivered B5-D3 engages airway phagocytes to promote early clearance and immune activation, which uncovers a distinct antiviral mechanism and offers new insights into the rational design of intranasal decoy-based interventions.

Results

Engineered sACE2-Fc decoys with two single mutations achieve robust neutralization against SARS-CoV-2 variants

To enable an optimal performance of the ACE2 decoy, we adopted the established sACE2-Fc fusion design [25] (Fig. 1a; Supplementary Fig. 1) and examined selected combinations of mutations that enhance the binding of human ACE2 to SARS-CoV-2 spikes [4] (B2–B6) or abolish enzymatic activity [11,12,33] (A2, A3, D1–D5) (Supplementary Fig. 2a,b). Indeed, mutants B2–B6 showed enhanced neutralization capacity in pseudovirus-based neutralization assays [9] against both Wuhan-Hu-1 and D614G pseudoviruses [19] (Fig. 1b; Supplementary Fig. 2c, d). Among these, the single mutation T92Q (B5), which increases spike affinity by removing a critical glycosylation site at N90 [40], demonstrated neutralization enhancement comparable to other multi-mutants. Mutations within the catalytic domain (A2, A3, D1, D3, D4, D5, but not D2), whether alone or in combinations, had minimal effect on spike binding but effectively abolished enzymatic activity (Supplementary Fig. 2e). We next combined T92Q with each of the inactivating mutations. Notably, the resulting double mutants, namely B5-D1, B5-D3, B5-D4, B5-D5, all retained strong neutralization while lacking enzymatic function (Fig. 1b; Supplementary Fig. 3a-c). Among these, B5-D3 (T92Q/H374N) emerged as the top candidate (Fig. 1a, red stars; Fig. 1b, red arrow), exhibiting both functional efficacy and structural integrity, with minimal deviation from wild type (WT) ACE2 in structural modeling (root mean square deviation [RMSD] = 0.212 Å; Supplementary Fig. 3d) [1].

To assess the breadth of neutralization, we tested the double mutants against pseudoviruses bearing spikes from various variants of concern (VOCs) and variants of interest (VOIs) [3,10,13,17,30,44]. All constructs showed dose-dependent neutralization with higher potency than WT sACE2-Fc (Fig. 1c, d; Supplementary Fig. 4). We further examined B5-D3 using plaque reduction neutralization tests (PRNTs) in Vero E6 cells, which indeed, confirmed its robust activity against authentic Wuhan-Hu-1, Delta, and Omicron variants BA.5, BQ.1.22, and XBB.1.5 [2,30,44,45] (Fig. 1e-g). In contrast, Casirivimab, serving as positive control [46], showed efficacy only against early variants (Wuhan-Hu-1 and Delta; Fig. 1f), but failed to neutralize Omicron sublineages (Fig. 1g). These results demonstrate that a rationally engineered sACE2-Fc decoy with only two mutations could achieve potent and safe neutralization across SARS-CoV-2 variants, reducing the potential risks associated with extensive mutagenesis.

Prophylactic administration of the sACE2-Fc B5-D3 mutant via the intranasal route exhibits superior protection against SARS-CoV-2

Next, we evaluated the *in vivo* efficacy of the sACE2-Fc double mutant B5-D3 against SARS-CoV-2 infection using aged K18-hACE2 mice (10–12 months old) (Fig. 2a). 6 hours (h) before inoculating with 1×10^4 plaque-forming unit (PFU) of SARS-CoV-2 (Wuhan-Hu-1 strain), mice received a prophylactic dose of recombinant B5-D3 protein either intranasally (IN, 2.5 mg/kg) or intravenously (IV, 15 mg/kg). To simulate a therapeutic intervention, an additional group received IV B5-D3 (15 mg/kg) 24 h post-virus inoculation. The vehicle control group received an intranasal PBS dose 6 h before viral challenge. Over a 14-day observation period, all mice in the PBS group exhibited significant weight loss and succumbed to infection by 7 days post-infection (dpi) (Fig. 2b, c, black lines). Both IV-treated groups exhibited initial weight loss similar to the PBS group; however, two out of four mice in each group began to regain weight from 10 dpi and survived until the observation endpoint (green and blue lines). Notably, all mice in the IN-prophylaxis group, despite receiving a 6-fold lower dose of B5-D3 protein, maintained stable body weight and achieved complete survival over the 14-day period (red lines).

To monitor viral burden, one mouse from each group was sacrificed at 4 dpi (Fig. 2a). No infectious viral particles were detected in lung homogenate from the IN-prophylaxis mouse. In contrast, mice treated with IV prophylaxis or therapy showed reduced but still detectable viral titers compared to the PBS group (Fig. 2d). Immunohistochemistry (IHC) staining further

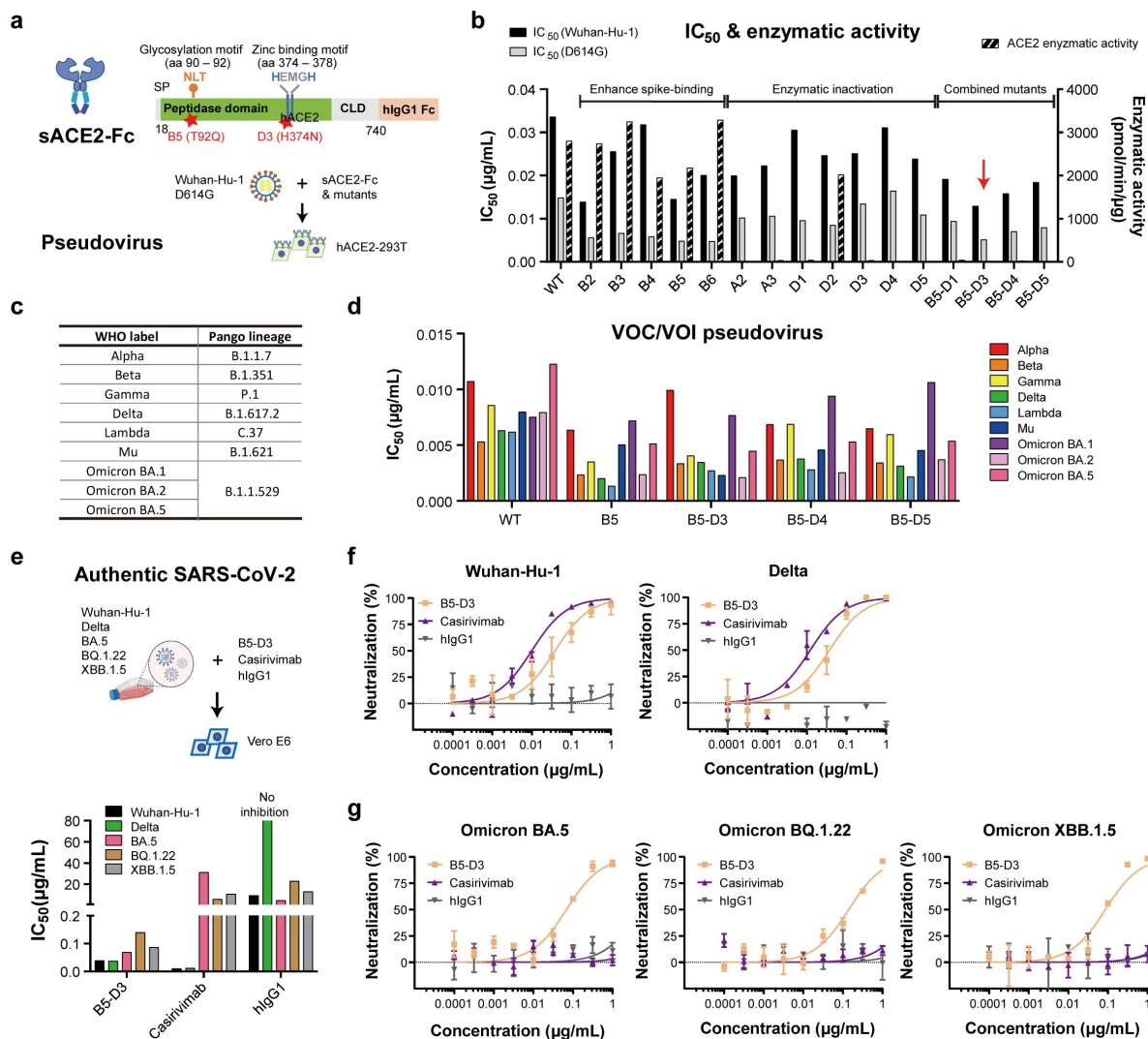


Fig. 1

Enhanced sACE2-Fc with two single mutations exhibited broad-spectrum neutralization of SARS-CoV-2 variants.

a Schematic representation of sACE2-Fc structure (upper) and neutralization assay setup (lower). Key amino acid positions (90-92 and 374-378) involved in glycosylation and zinc binding are highlighted. Red stars mark the positions of mutations in the sACE2-Fc mutant B5-D3. SP, signal peptide; CLD, collectrin-like domain; hlgG1, human IgG1. **b** Comparative bar graph showing the half-maximal inhibitory concentration (IC_{50}) values for neutralization of Wuhan-Hu-1 and D614G pseudoviruses by WT sACE2-Fc and mutants (B2 to B6, A2, A3, D1 to D5, and B5-derivatives). The red arrow emphasizes the superior performance of the B5-D3 mutant. Enzymatic activity of each construct is plotted on the right axis. **c** List of pseudoviruses carrying spikes from different SARS-CoV-2 variants tested, categorized by the World Health Organization (WHO) into VOCs and VOIs. **d** Graph displaying IC_{50} values of WT sACE2-Fc, B5, and B5-D3/4/5 mutants against various SARS-CoV-2 VOCs and VOIs in neutralization assays. **e** Schematics of the plaque-reduction neutralization tests (PRNTs) process (upper) and the resulting IC_{50} values for B5-D3, Casirivimab, and hlgG1 against authentic SARS-CoV-2 (lower). **f, g** Dose-response curves depicting the neutralization efficacy of B5-D3 (orange), Casirivimab (purple), and hlgG1 (grey) in PRNTs against authentic SARS-CoV-2 Wuhan-Hu-1 and Delta strains (**f**), and Omicron sub-lineages (**g**). Data are presented as mean $\pm$ standard deviation (SD) from duplicate experiments.

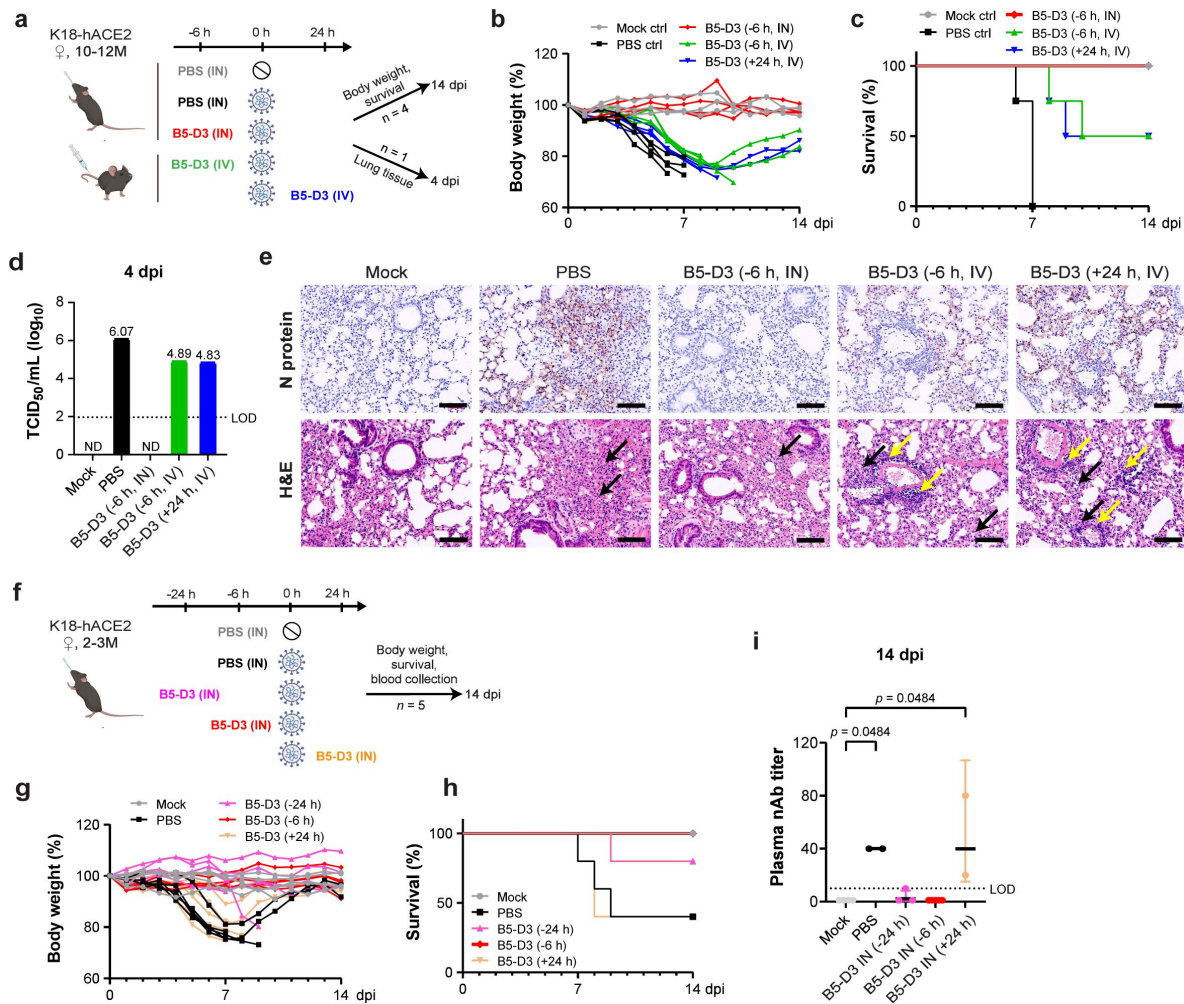


Fig. 2

Enhanced survival and reduced infection in K18-hACE2 mice through intranasal prophylaxis with B5-D3 against SARS-CoV-2.

a-e Female K18-hACE2 mice, aged 10 to 12 months, were inoculated with 1×10^4 PFU of SARS-CoV-2 (Wuhan-Hu-1 strain). Mice were treated with B5-D3 6 h prior (-6 h) via intranasal (IN, red) or intravenous (IV, green) routes, or 24 h post-infection (+24 h, blue) via IV ($n = 5$ each). IN PBS administered 6 h prior to viral challenge served as the vehicle control (black; $n = 5$), and PBS alone was used for mock control (grey; $n = 4$) (**a**). Body weight and survival were monitored over 14 days (**b**, **c**). One mouse from each group was sacrificed at 4 dpi for analysis of viral titers in lung homogenates using a median tissue culture infectious dose (TCID₅₀) assay (**d**). IHC staining for N protein and H&E staining were performed on lung tissues collected at 4 dpi (**e**). Black arrows indicate alveolar thickening, and yellow arrows show leukocyte infiltration. Scale bar = 100 µm. ND, not detected; LOD, limit of detection. **f-i** Young female K18-hACE2 mice, aged 2 to 3 months, were inoculated similarly and treated with B5-D3 via IN route at 24 h before (-24 h, pink), 6 h before (-6 h, red), or 24 h after (+24 h, orange) the viral challenge ($n = 5$). Mice receiving IN PBS 6 h before infection served as the vehicle control (black), with mock control mice receiving PBS alone (grey) (**f**). Body weight (**g**) and survival (**h**) were recorded for 14 days. Neutralizing antibody titers against Wuhan-Hu-1 in serum samples from surviving mice at 14 dpi were determined using Vero E6 cells (**i**). nAb, neutralizing antibody. Data are presented as the geometric mean $\pm$ geometric SD. Statistical significance was determined using Dunn's multiple comparisons test.

confirmed the absence of viral nucleocapsid (N) protein in the IN-treated mouse, whereas IV-treated mice showed residual infection and immune cell infiltration. Hematoxylin and eosin (H&E) staining revealed varying degrees of alveolar thickening in all mice (**Fig. 2e** [↗](#)).

To further explore the timing of IN administration, we treated a younger cohort of K18-hACE2 mice (2 – 3 months old) with B5-D3 (IN, 2.5 mg/kg) at –24 h, –6 h, or +24 h relative to viral challenge (**Fig. 2f** [↗](#)). Similarly, the PBS group exhibited substantial weight reduction from 4 dpi and reached approximately 20% loss by 7 dpi (**Fig. 2g** [↗](#), black lines). Despite this, two of the five infected young mice eventually recovered, resulting in 40% survival (**Fig. 2g, h** [↗](#), black lines). Interestingly, both the –24 h and –6 h IN prophylaxis groups maintained stable body weights (**Fig. 2g** [↗](#), pink and red lines), resulting in survival rates of 80% and 100%, respectively (**Fig. 2h** [↗](#)). In contrast, the +24 h group showed substantial weight loss and no survival improvement compared to the PBS group, indicating limited therapeutic benefit when IN B5-D3 was administered post-infection (**Fig. 2g, h** [↗](#), orange lines). Consistently, virus-neutralizing antibodies were detected in surviving mice from the PBS and +24 h groups at 14 dpi, indicating active infection and subsequent immune response. Whereas antibody levels remained minimal in the two IN-prophylaxis groups, suggesting effective prevention of viral replication (**Fig. 2i** [↗](#)).

Efficient protection against SARS-CoV-2 by intranasal B5-D3 prophylaxis depends on Fc-mediated effector functions

Prompted by the significant protection conferred by IN prophylaxis with B5-D3, we examined the early responses following SARS-CoV-2 challenge in K18-hACE2 mice. A new cohort of 2- to 3-month-old mice received B5-D3 IN treatment 6 h before infection (–6 h), and lung tissues were harvested at 1, 2, and 4 dpi for analysis (**Fig. 3a** [↗](#)). An additional group was treated with a modified version of B5-D3, which contains L234A/L235A mutations in the human IgG1 Fc region (B5-D3-LALA) to abrogate Fc effector functions [28 [↗](#)] (Supplementary Fig. 5). Quantitative PCR of viral spike (S) and nucleocapsid (N) RNA in lung tissues revealed only marginal viral loads in the B5-D3-treated mice as early as 1 dpi, indicating efficient suppression of early viral replication compared to the PBS group (**Fig. 3b** [↗](#)). Analysis of infectious viral particles in lung homogenates further corroborated these observations, demonstrating minimal or undetectable viral titers in the B5-D3 group at all time points (**Fig. 3c** [↗](#)). In contrast, PBS-treated mice exhibited consistently high viral loads. Interestingly, the B5-D3-LALA group displayed varied outcomes, with significant viral burdens observed in two out of three mice, indicating that Fc effector functions are critical for B5-D3-mediated protection (**Fig. 3b, c** [↗](#), right panels). Consistently, IHC staining for N protein in lung sections confirmed the absence of viral infection in the B5-D3 group at all time points. Whereas signs of viral replication were evident in the lungs of mice treated with PBS at as early as 1 dpi and in the B5-D3-LALA-treated cohort by 4 dpi (**Fig. 3d** [↗](#); Supplementary Fig. 6, left panels). Despite variations in viral burden, H&E staining indicated alveolar thickening in all groups. Particularly, the alveolar changes observed in the absence of detectable infection in B5-D3–treated mice suggested immune activation without viral replication (**Fig. 3e** [↗](#); Supplementary Fig. 6, right panels). These findings demonstrated that IN prophylaxis with B5-D3 blocks SARS-CoV-2 infection not only by neutralization but also by immune mechanisms such as Fc-mediated effector functions.

RNA-Seq analysis of lung transcriptomes reveals early antigen presentation and prompt viral clearance following SARS-CoV-2 neutralization by B5-D3

To delineate the immune mechanisms underlying B5-D3-mediated prophylactic protection against SARS-CoV-2, we examined the transcriptomes of lung samples collected at 1, 2, and 4 dpi from the above experiment (**Fig. 4a–d** [↗](#); Supplementary Fig. 7a). Unsupervised clustering based on Pearson correlation distinguished samples with severe infection (mainly PBS-treated) from those with

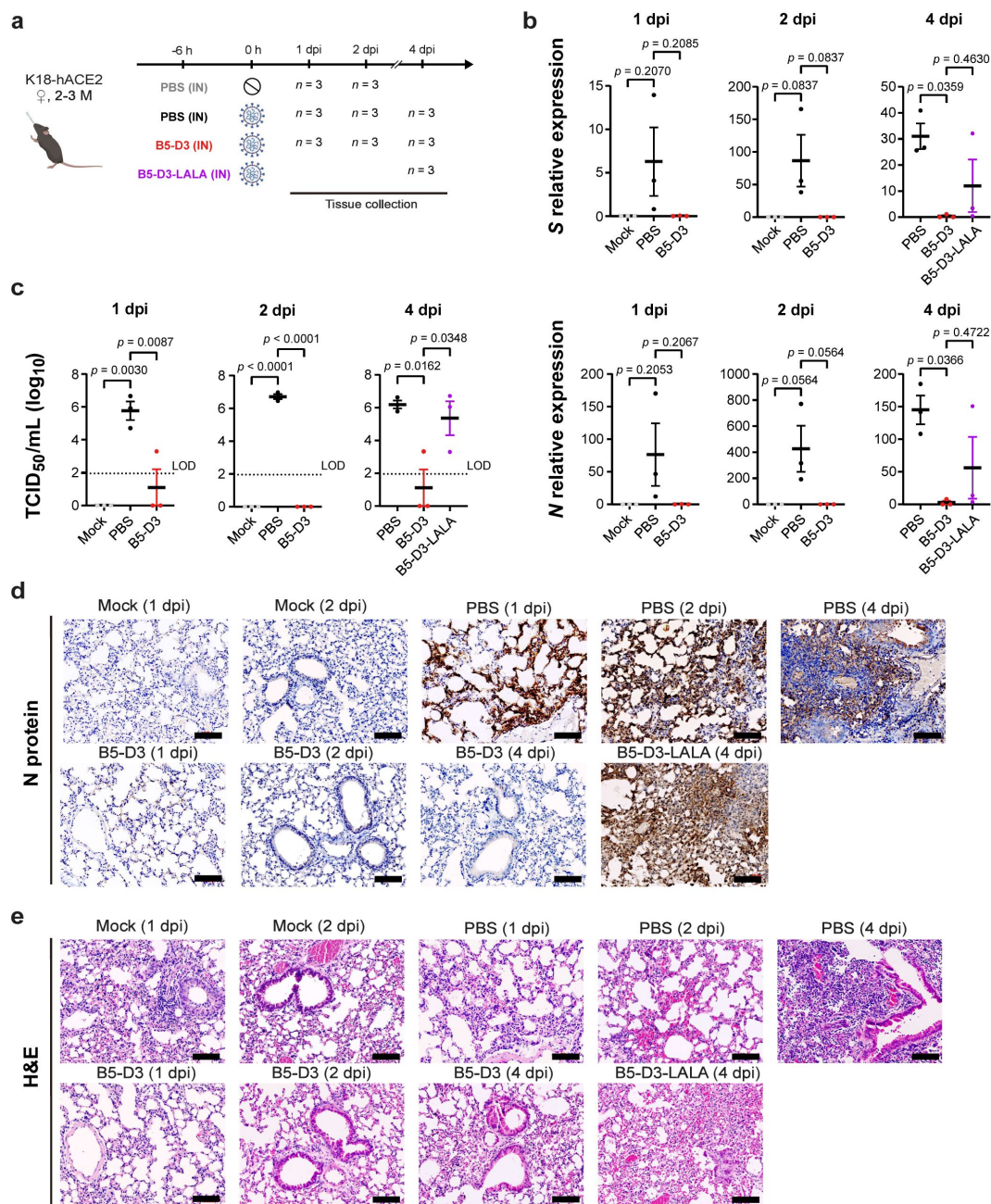


Fig. 3

Efficient viral clearance at early stages through intranasal prophylaxis with B5-D3 against SARS-CoV-2 challenge in K18-hACE2 mice.

a Workflow diagram showing timelines and treatments for different mouse groups. Young female K18-hACE2 mice aged 2 to 3 months received prophylactic administration of PBS (black), B5-D3 (red), or B5-D3-LALA (purple) via the IN route 6 h prior to inoculation with 1×10^4 PFU of Wuhan-Hu-1. Mice inoculated with PBS instead of the virus served as mock controls (grey). Mice from each treatment group were sacrificed for tissue collection at 1, 2, and 4 dpi ($n = 3$ per time point). **b** Quantitative PCR results showing relative amounts of *S* (upper) and *N* (lower) viral RNA in lung tissues collected from different groups at 1, 2, and 4 dpi, normalized to mouse *Gapdh* (a). **c** The titers of infectious viruses detected in lung homogenates, measured by TCID₅₀ assays at 1, 2, and 4 dpi. **d, e** Fixed lung tissues were sectioned and stained; IHC for viral N protein (d) and H&E staining for tissue damage (e) are shown (scale bar = 100 μ m). Data presented as mean $\pm$ standard error of the mean (SEM). Statistical significance was determined by Tukey's multiple comparisons test.

subtle or no infection (mocks and most decoy-treated mice) (Supplementary Fig. 7a). Corroborating the levels of viral infections observed, differential gene expression (DGE) analysis revealed extensive inflammatory responses in the PBS groups, significantly greater than in mock treatments. At 1, 2, and 4 dpi, 26, 1232, and 1756 genes were upregulated, respectively, and were significantly enriched in Gene Ontology Biological Process (GOBP) terms related to antiviral responses such as type I interferon (IFN) responses and innate immune responses (**Fig. 4a, b**; Supplementary Fig. 7b-d) [47]. In stark contrast, DGE analysis between B5-D3 prophylaxis and mocks at 1, 2, and 4 dpi showed subtle changes, with only 1, 7, and 32 genes upregulated, respectively, and only moderate enrichment in chemotaxis-related pathways at 4 dpi (**Fig. 4c**; Supplementary Fig. 7e). The B5-D3-LALA group, however, had 264 genes upregulated at 4 dpi compared to the mocks, suggesting incomplete protection and ongoing viral activity (**Fig. 4d**; Supplementary Fig. 7f).

To capture the immune activations specifically linked to B5-D3-triggered antiviral efficacy other than infection-induced inflammation, we directly compared the B5-D3 and PBS groups (**Fig. 4e-j**; Supplementary Fig. 8, 9). Interestingly, at 1 dpi, the B5-D3 group exhibited enhanced expression of several immune-related genes, including *Lef1* [37], *Fscn1* [49], *Kcne4* [8], *Tcrb*, and *Ccl22* [20, 35], which are associated with early dendritic cell function and T cell activation (**Fig. 4e**). Gene Set Enrichment Analysis (GSEA) of GOBPs and Kyoto Encyclopedia of Genes and Genomes (KEGG) pathways further supported these findings. Chemotaxis and pathways related to antigen presentation such as Rap1 signaling pathway [16] and Th1 and Th2 cell differentiation were significantly activated in the B5-D3 group at 1 dpi compared to PBS group (**Fig. 4h-j**; Supplementary Fig. 8a-c). Moreover, the B5-D3 groups showed enhancement in cilium movement and metabolism of xenobiotics at both 2 and 4 dpi, suggesting active clearance of viral particles due to effective early responses (Supplementary Fig. 9c-f).

Furthermore, we collectively examined the GOBPs that were significantly activated in B5-D3 groups at either 1, 2, and 4 dpi among all treatment groups and time points. Markedly, B5-D3 group showed higher normalized enrichment scores (NES) in chemotaxis-related GOBP pathways than PBS group at 1 dpi (**Fig. 4k**, purple boxes), while direct comparison between B5-D3 and PBS groups further revealed the broad involvement of multiple types of effector immune cells (**Fig. 4l**, purple box). These results collectively indicate that early immune activation is a hallmark of B5-D3-mediated protection.

Finally, the lung transcriptomes from mice receiving B5-D3 without viral inoculation showed high similarity to the PBS vehicle controls (Supplementary Fig. 10a). The 10 upregulated genes identified showed poor correlation with the virus-inoculated B5-D3 group (Supplementary Fig. 10b, c), supporting that early immune responses observed in B5-D3 IN prophylaxis groups were primarily triggered by virus neutralization rather than by B5-D3 alone.

Intranasally delivered B5-D3 is enriched in the respiratory tract and targets mainly the airway macrophages

The superior antiviral effects of IN over IV administration of B5-D3 as observed in the K18-hACE2 infection experiments suggested the importance of mobilizing the local immunity within the respiratory tract. To track IN B5-D3, we labeled B5-D3 protein with Alexa Fluor 750 (AF750) and examined its bio-distribution and kinetics after IN administration (**Fig. 5a**). *In vivo* imaging showed that fluorescence-labeled B5-D3 (B5-D3-AF750) was present in the nasal cavities for at least 24 h after a single IN dose in K18-hACE2 mice (**Fig. 5b**). *Ex vivo* images further revealed that B5-D3-AF750 distributed in the respiratory tract from nasal cavity to lung within 20 min and remained enriched in lungs by 24 h after administration (**Fig. 5c**). In contrast, non-respiratory organs showed minimal signals, which were merely detectable in urinary system and liver (**Fig. 5d**).

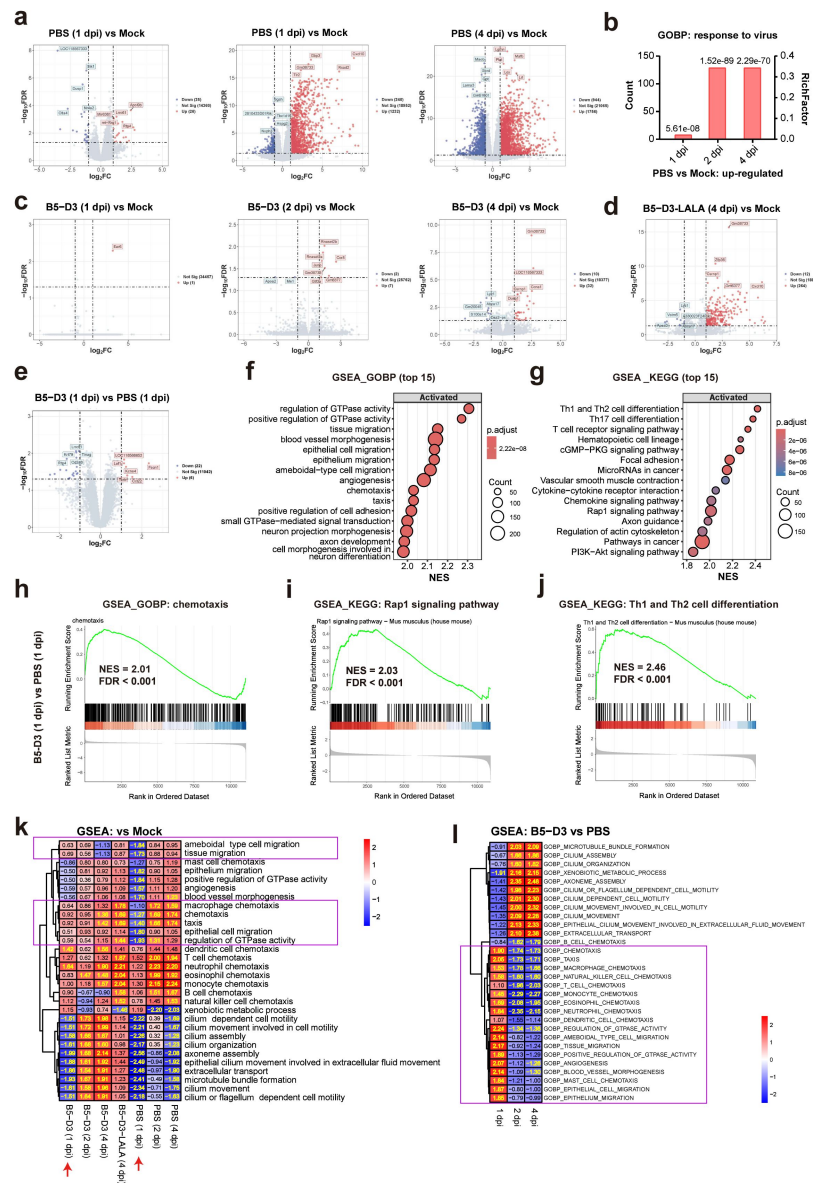


Fig. 4

Transcriptomic analysis revealed early immune activation in IN B5-D3-prophylaxis mouse group after SARS-CoV-2 challenge.

a-d DGE analysis comparing PBS (**a**), B5-D3 (**c**), and B5-D3-LALA (**d**) against the mock control at specific time points ($n = 3$). Volcano plots illustrate the gene expression changes (**a, c, d**), while red and blue dots represent significantly upregulated and downregulated genes, respectively, with $|\log_2 \text{fold change} (\log_2 \text{FC})| \geq 1$ and a false discovery rate (FDR) < 0.05 . Bar chart in **b** shows the enrichment of GOBP “response to virus” observed in PBS groups at 1, 2, 4 dpi, in which adjusted p values are indicated for individual comparisons. **e-g** Comparison between IN B5-D3 and PBS group at 1 dpi. Volcano plot illustrates the DGE analysis between IN B5-D3 to PBS group at 1 dpi (**e**), with red and blue dots representing significantly upregulated and downregulated genes, respectively, with $|\log_2 \text{FC}| \geq 1$ and FDR < 0.05 . GSEA shows top 15 significantly activated GOBPs (**f**) and KEGG pathways (**g**) in IN B5-D3 compared to PBS group at 1 dpi. NES, normalized enrichment score; p.adj, adjusted p value. **h-j** GSEA plots of chemotaxis (**h**), Rap1 signaling pathway (**i**), and Th1 and Th2 cell differentiation in B5-D3 vs PBS comparison at 1 dpi. **k, l** Heatmaps show NES of GSEA comparing various treatments to the mock control (**k**) and between B5-D3 to PBS (**l**), focusing on top 10 GOBPs highlighted in **f** and **Supplementary Figure 9c, d**, respectively, and those related to immune cell chemotaxis. Significant NES values ($p < 0.05$, FDR < 0.25) were highlighted in yellow. Benjamin-Hochberg method was used for FDR adjustment.

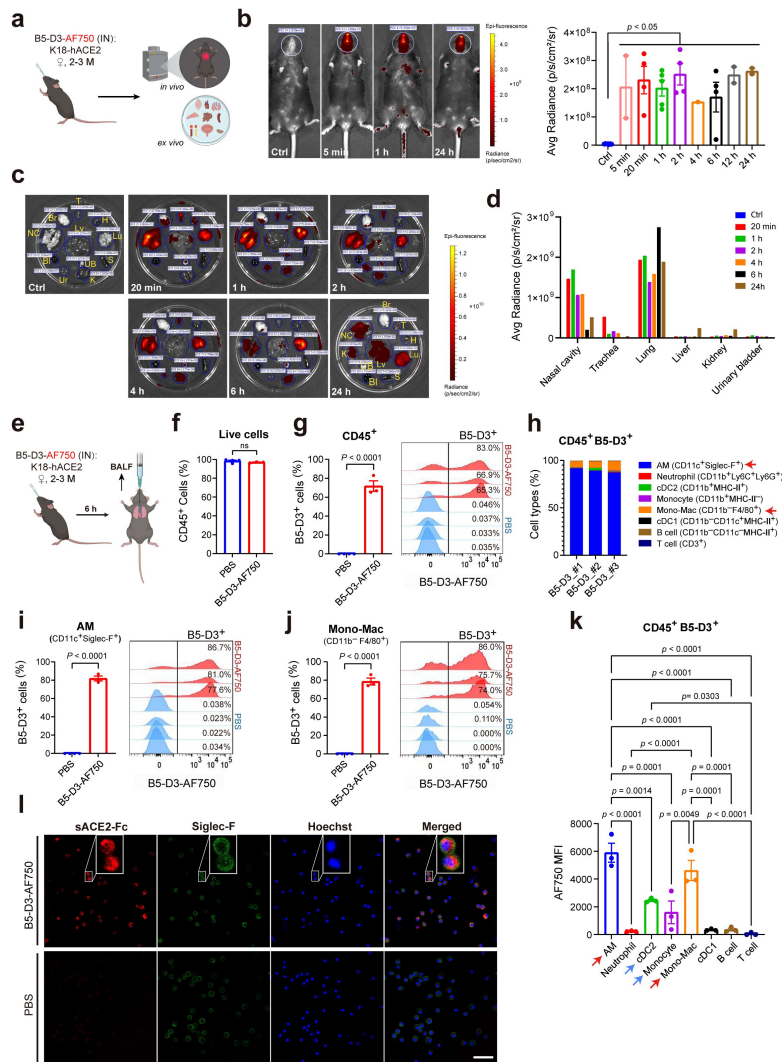


Fig. 5

***In vivo* bio-distribution of B5-D3 after IN administration.**

a Schematic workflow of *in vivo* and *ex vivo* imaging. Female K18-hACE2 mice aged 2 to 3 months received IN administration of fluorescently labeled B5-D3 (B5-D3-AF750) and was visualized at different time points. **b** Representative whole-body images of control and treated mice at 5 min, 1 h, and 24 h after B5-D3-AF750 administration, showing the signal captured by *in vivo* imaging (left). White circles indicate regions of interest (ROIs) for quantification of fluorescence signals in the nasal cavities. Average (Avg) Radiance measured at all time points are shown on the right. **c** *Ex vivo* images of tissues from control and treated mice sacrificed at indicated time points after B5-D3-AF750 administration. Blue circles indicate ROIs for signal quantification. Br, brain; NC, nasal cavity; T, trachea; Lu, lung; H, heart; Lv, liver; S, spleen; K, kidney; UB, urinary bladder; Bl, blood; Ur, urine. **d** Avg Radiance shows the fluorescence signals in excised tissues measured *ex vivo*. **e** Schematic workflow for BALF analysis. Female K18-hACE2 mice aged 2 to 3 months received IN administration of B5-D3-AF750 ($n = 3$) or PBS ($n = 4$) and were sacrificed at 6 h later for collection of BALF cells. **f** Percentage of CD45⁺ cells in live BALF cells. **g** Positive rates (left) and histograms (right) of B5-D3 binding/uptake in CD45⁺ BALF cells. Histograms show B5-D3-AF750 fluorescence intensities in CD45⁺ BALF cells from individual mice. **h** Frequency of individual immune cell types in CD45⁺B5-D3⁺ BALF cells. Red arrows point out AM and Mono-Mac with high abundance. AM, alveolar macrophage; Mono-Mac, monocyte-derived macrophage; cDC1/2, type 1 or 2 conventional dendritic cells. **i, j** Positive rates (left) and histograms (right) of B5-D3 binding/uptake in CD11c⁺Siglec-F⁺ AMs (**i**) and CD11b⁺F4/80⁺ mono-Macs (**j**). **k** Median fluorescence intensity (MFI) of AF750 indicate B5-D3 binding/uptake in different CD45⁺B5-D3⁺ populations. **l** Confocal images (scale bar = 50 μ m) of BALF cells collected at 6 h and stained for sACE2-Fc (red, Abcam, ab98596), Siglec-F (green, BD #564514), and nuclei (blue, Hoechst). Magnified views are shown in white rectangles. Data are presented as mean $\pm$ SEM, and statistical significance was determined by Tukey's multiple comparisons test or Student's t-test.

To further identify the immune cells in the respiratory tract that are actively engaged with IN B5-D3, we performed flow cytometry analysis on bronchoalveolar lavage fluid (BALF) at 6 h after IN administration of B5-D3-AF750 (**Fig. 5e**; Supplementary Fig. 11a). Corroborating previous findings, over 95% of live BALF cells were CD45⁺ immune cells, predominantly CD11c⁺Siglec-F⁺ resident alveolar macrophages (AMs) in both treatment and vehicle groups (**Fig. 5f**; Supplementary Fig. 11b). Notably, the IN administered B5-D3-AF750 was actively retained in the CD45⁺ cells, with positive rates exceeding 65% in all treated mice (**Fig. 5g**). Among the CD45⁺B5-D3⁺ cells, more than 95% were macrophages, composed primarily of CD11c⁺Siglec-F⁺ AMs (87.2 – 91.7%) and Siglec-F⁺CD11b⁺F4/80⁺ monocyte-derived macrophages (mono-Macs; 6.6 – 9.9%) (**Fig. 5h**, red arrows). Consistently, these macrophage populations also exhibited the highest B5-D3 positive rates (**Fig. 5i, j**; Supplementary Fig. 12) and greatest median fluorescent intensities (MFI) (**Fig. 5k**, red arrows) among all immune cell types in the BALF, indicating the strongest B5-D3-AF750 uptake. Other phagocytic cell types such as the type 2 conventional dendritic cells (cDC2) and monocytes also exhibited considerable AF750 intensities (**Fig. 5k**, blue arrows; Supplementary Fig. 12b, c), suggesting potential relationships between B5-D3 uptake and phagocytic activities. Confocal microscopy of BALF cells after immunostaining further confirmed that the B5-D3-AF750 were present in the cytoplasm after being retained in AMs (**Fig. 5l**). Together, these results demonstrate that intranasal B5-D3 preferentially accumulates in the respiratory tract and is predominantly taken up by airway macrophages, supporting their important role in mediating early immune responses.

sACE2-Fc facilitates phagocytosis of SARS-CoV-2 pseudovirus via mechanisms distinct from ACE2-dependent viral infection

To examine the implication of macrophage involvement in the early immune activation observed in IN B5-D3 treatment groups, we performed cellular analysis using THP-1 cells as an *in vitro* model for phagocytes [5] and examined the sACE2-Fc-dependent phagocytosis of spike-pseudotyped lentiviruses. Indeed, immunostaining of HIV capsid protein p24 confirmed the attachment and entry of pseudoviruses in the THP-1 cells in a B5-D3-dependent manner, with an evidenced signal peak at 6 h post-co-incubation (Supplementary Fig. 13). Interestingly, analysis on the THP-1-derived M0 and M1 macrophages detected even greater p24 signals, indicating stronger phagocytosis activities compared to undifferentiated THP-1 cells (**Fig. 6a, b**; Supplementary Fig. 14a, b, d, e). This process resembled antibody-dependent cellular phagocytosis (ADCP), which was significant in THP-1-derived M0 and M1 macrophages [42]. Consistently, further examination revealed colocalization of internalized pseudovirus with lysosomal associated membrane protein 1 (LAMP1), suggesting trafficking to lysosomes for degradation [41] (**Fig. 6c**; Supplementary Fig. 14c, f).

We further compared this finding with the pseudovirus uptake in Calu-3 cells overexpressing human ACE2 (hACE2-Cal-3; Supplementary Fig. 15) as a model of lung epithelial cells. Distinctly, co-incubation with B5-D3 significantly reduced the pseudovirus entry in hACE2-Cal-3 (**Fig. 6d-f**). Interestingly, despite the evident pseudovirus uptake facilitated by B5-D3 in the THP-1 and derivative macrophages, there was no correspondingly detectable luciferase activity in these cells, which indicates viral degradation within phagolysosomes instead of viral genome release or transgene expression (**Fig. 6g**). Corroborating these observations, western blot analysis showed absence of cleaved S2' fragments in sACE2-Fc-treated macrophages, supporting that the pseudoviruses did not undergo membrane fusion or cytosolic release following uptake [50] (**Fig. 6h**). Collectively, these findings suggest that IN B5-D3 not only blocks viral entry into epithelial cells but also actively redirects SARS-CoV-2 to phagocytic clearance by engaging airway phagocytes via Fc-dependent mechanisms. Moreover, such ADCP-like process likely contributes to early immune activation and restricts the infection at the respiratory mucosal surface.

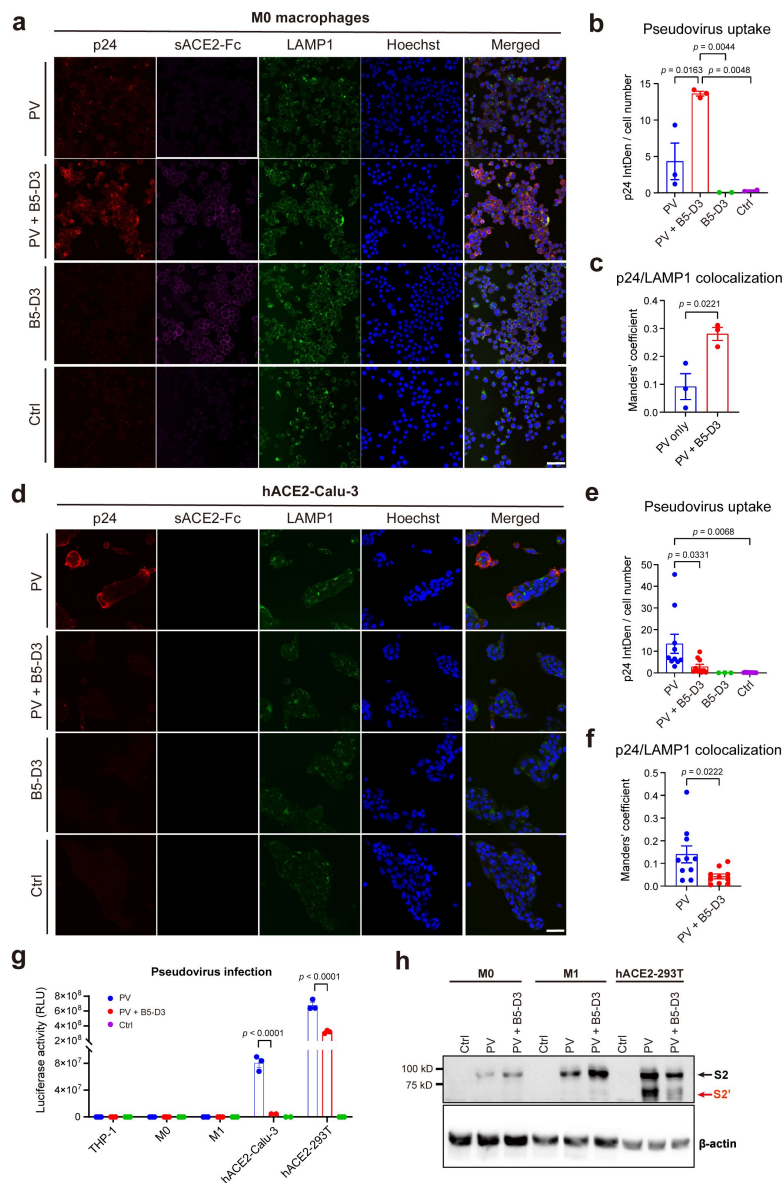


Fig. 6

B5-D3 enhanced phagocytosis and degradation of SARS-CoV-2 pseudovirus in THP-1-derived macrophages.

a Immunostaining of p24 (Invitrogen #PA5-81773), sACE2-Fc (Abcam #ab98596), and LAMP1 (Abcam #ab25630) in THP-1-differentiated M0 macrophages showing phagocytosis of SARS-CoV-2 pseudovirus (p24⁺) after 6 h of incubation with or without sACE2-Fc (scale bar = 50 μm). LAMP1 was stained to identify lysosomes. **b** Quantification of p24 signal intensity as shown in **a**. Intensity Density (IntDen) per cell number indicates the mean p24 signal per cell, calculated using ImageJ. Each dot represents one image. **c** Manders' coefficient indicating the colocalization of p24 and LAMP1 in THP-1 M0 macrophages as shown in **a**. **d** Immunostaining of p24, sACE2-Fc, and LAMP1 in hACE2-Calu-3 cells after 6 h incubation with pseudovirus, with or without B5-D3 (scale bar = 50 μm). **e** Quantification of mean p24 signal intensity as shown in **d**. **f** Manders' coefficient for the colocalization of p24 and LAMP1 in hACE2-Calu-3 cells, as shown in **d**. **g** Quantification of pseudovirus infection in THP-1, M0 macrophages, M1 macrophages, hACE2-Calu-3, and hACE2-293T cells, in the presence or absence of sACE2-Fc. Results shown were luciferase activities measured at 2 days post-transduction. **h** Immunoblot staining to detect SARS-CoV-2 spike cleavage after cell entry. M0 macrophages, M1 macrophages, and hACE2-293T cells were incubated with pseudovirus for 6 h, with or without sACE2-Fc, before protein extraction. Data are presented as mean ± SEM, and statistical significance was determined by Tukey's multiple comparisons test.

Discussion

In this study, we comprehensively evaluated the protective efficacy and mechanistic basis of an optimized sACE2-Fc decoy (B5-D3) against SARS-CoV-2 infection. By introducing only two mutations (T92Q and H374N), we generated a minimally engineered sACE2-Fc mutant (B5-D3) that achieved broad-spectrum neutralization with minimal risk of disrupting the RAS. Among various administration routes and dosing schedules examined, we demonstrated that IN prophylaxis of B5-D3 achieved the most robust protection, completely preventing disease in both young and aged K18-hACE2 mice. Transcriptomics analysis of the infected lung samples at early time points revealed distinct IN B5-D3-dependent immune activation at the onset of infection, indicating B5-D3 acted not only as a viral decoy but also as an immune modulator. Corroborating these findings, bio-distribution analysis of fluorescence-labeled B5-D3 demonstrated rapid uptake and high accumulation in the respiratory tract, primarily within airway macrophages. Furthermore, phagocytosis assays supported that sACE2-Fc decoy mediated a rapid viral clearance in macrophages, while abolishing membrane ACE2-mediated infection in epithelial cells. Together, these findings reveal a dual-function mechanism for sACE2-Fc decoys in redirecting SARS-CoV-2 to phagocytic clearance and rapid immune engagement, supporting their potential as intranasal prophylactics against respiratory viruses.

Previously engineered sACE2-Fc [7,11] have achieved high neutralization potency against SARS-CoV-2 by incorporating up to five mutations. However, the extensive mutagenesis raises concerns about structural instability [11], reduced production efficiency [4], and potential immunogenicity [43]. In this study, we adopted a minimalistic approach by introducing only two targeted mutations in sACE2-Fc. The optimized B5-D3 retains a conformation closely resembling native ACE2 and exhibits robust viral neutralization and enzymatic inactivation. These results underscore the feasibility of achieving optimal antiviral potency while preserving protein stability and safety through limited mutagenesis.

A major challenge in controlling SARS-CoV-2 and other respiratory virus infections is that systematically infused monoclonal antibodies and vaccine-induced immunity primarily target viruses in the circulation, which may only represent a minor fraction of viral burden during early infection. This limits their ability to respond to initial viral infection in respiratory epithelium to prevent massive propagation. Our study and others' work [14] have consistently shown that systemic administration of sACE2-Fc decoy or monoclonal antibodies, either before or after the virus inoculation, exhibited limited efficacy against virus replication and pathogenesis (**Fig. 2b**, blue and green lines). In contrast, prophylaxis with IN B5-D3 resulted in complete protection, with 100% survival and no detectable weight loss in the -6 h group (**Fig. 2b, g**, red lines). Notably, this protection was observed in both young and aged mice, which otherwise exhibited 60% and 100% mortality, respectively, following infection (**Fig. 2c, h**, black lines). These findings underscore the importance of rapid antiviral action at the site of viral entry, which is particularly crucial to protect against respiratory viruses.

IN prophylaxis with monoclonal antibodies or other sACE2Fc mutants have been shown to be effective against SARS-CoV-2 [14,18,21], and studies have found that sACE2-Fc protects against virus through the Fc-effector functions [7]. However, the precise mechanism has not been depicted, which prevents the further development of these approaches for translation. Here, our study provided evidence for a deeper mechanistic insight, showing that IN sACE2-Fc decoys rapidly engage host immunity in the respiratory tract. Consistently with others' work [7], IN delivery of a Fc-null variant (B5-D3-LALA) resulted in suboptimal protection and higher viral infection compared to B5-D3. Notably, despite the minimal infection observed, B5-D3-treated mice showed robust early immune activation, including induction of antigen presentation and T cell

activation within 24 h post-infection (**Fig 4**). These support that B5-D3 not only neutralizes virus but also primes innate and adaptive immune responses, counteracting early-stage viral immune evasion.

Furthermore, our bio-distribution data showed that IN-delivered B5-D3 preferentially accumulates in the respiratory tract (**Fig. 5a–d**). Flow cytometry and confocal imaging confirmed strong binding and uptake of B5-D3 by airway phagocytes, primarily alveolar macrophages and monocyte-derived macrophages (**Fig. 5e–l**). Notably, phagocytosis assays demonstrated that B5-D3–virus complexes were trafficked to lysosomes for degradation in macrophages. These findings support a mechanism in which B5-D3 redirects viral particles away from membrane ACE2-dependent epithelial entry and toward phagocytic clearance (**Fig. 7**). Importantly, this ADCP-like process likely also facilitates early activation of pattern recognition receptors (PRRs) and initiates downstream antiviral signaling cascades before the virus reaches epithelial targets (**Fig. 7b**). Hence, the IN prophylaxis offers a unique advantage by enabling localized immune priming and efficient viral clearance at the frontline of infection.

In conclusion, we present a rationally designed sACE2-Fc decoy with minimal mutagenesis (B5-D3) and provide compelling evidence and insights into the immune mechanism supporting its potent prophylactic efficacy. Intranasal prophylactic administration of B5-D3 not only neutralizes SARS-CoV-2 but also redirects the virus toward phagocytic clearance, enabling early immune engagement and complete protection. These findings provide a mechanistic basis for decoy-based antiviral strategies and offer a promising approach to combat current and future airborne viral threats. Further studies may aim to develop approaches to enhance the rapid local immune engagement to restrict early viral propagation. Additionally, regimen refinements are needed to enhance stability and functionality of decoy-based treatments before their clinical translation and extension to a broader range of respiratory pathogens.

Materials and methods

Plasmid construction

The coding sequence of human ACE2 was cloned into the pGEM-T easy vector (Promega) and underwent site-directed mutagenesis [4,11,23,33]. The sACE2 and human IgG1 hinge-Fc regions (amino acid [aa] 216–447) were assembled via overlapping PCR. These constructs, along with 6xHis-tagged versions, were inserted into the HDM-SARS2-Spike-delta21 vector (Addgene #155130) to generate HDM-CMV-sACE2(Fc)-his plasmids. L234A/L235A (LALA) in hIgG1 were introduced to generate HDM-CMV-sACE2-Fc-LALA-his plasmids. SARS-CoV-2 spike variants with or without an HA tag fused to the C-terminal were synthesized and inserted into the HDM vector [38].

Protein structure visualization

The crystal structure of SARS-CoV-2 spike receptor-binding domain bound with ACE2 (6M0J) was downloaded from the Protein Data Bank (PDB, <https://www.rcsb.org/structure/6m0j>) [22]. Color-labeling of individual amino acids was performed on the PDB website. For structural overlapping analysis of wild type (WT) sACE2 and B5-D3 (aa 18–740), protein structures were predicted using online AlphaFold 3 server (<https://alphafoldserver.com/>) [1]. PyMOL was utilized for root mean square deviation (RMSD) calculations and structural visualization.

Cell culture

293T, Vero E6, Calu-3, and THP-1 cells were obtained from the American Type Culture Collection and incubated at 37 °C with 5% CO₂. Specifically, 293T, Vero E6, and Calu-3 cells were maintained in Dulbecco's Modified Eagle Medium (DMEM, Gibco) supplemented with 10% fetal bovine serum

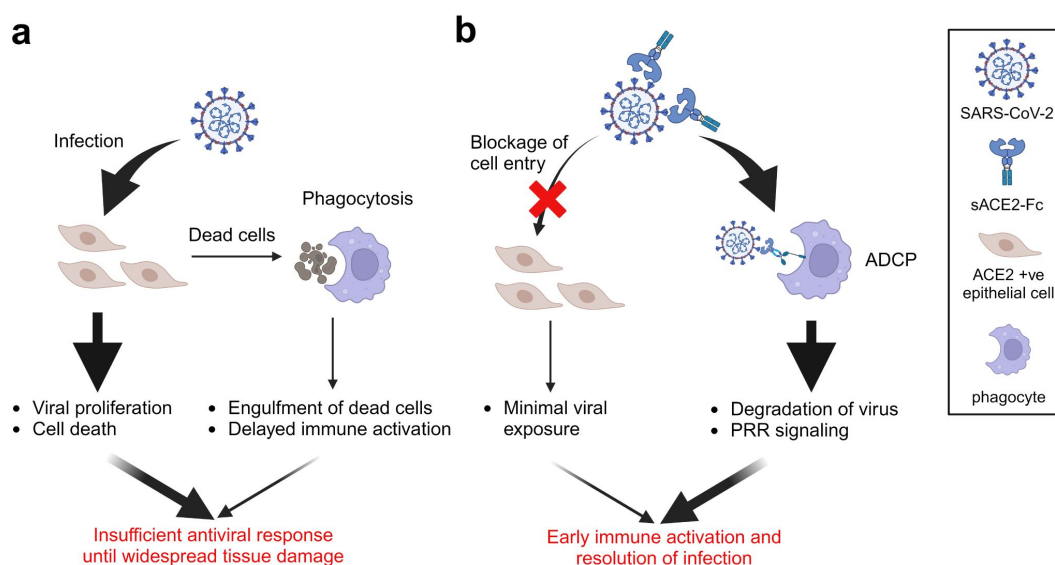


Fig. 7

Proposed mechanisms of action of IN sACE2-Fc decoy in preventing SARS-CoV-2 infection.

a, b Schematics illustrating the actions and outcomes of SARS-CoV-2 infection, in the absence (**a**) and presence (**b**) of IN delivered sACE2-Fc decoys. The figure was created with [BioRender.com](https://www.biorender.com).

(FBS, Gibco) and 1% Penicillin-Streptomycin (PS, Gibco). THP-1 cells were cultured in Roswell Park Memorial Institute 1640 medium (RPMI, Gibco) with similar supplements. THP-1 cells were differentiated into M0 macrophages using 50 nM phorbol 12-myristate 13-acetate (PMA) for 48 h, followed by a 24 h rest. For M1 macrophage differentiation, post-PMA treatment cells were stimulated with 10 ng/mL lipopolysaccharide and 20 ng/mL interferon (IFN)- γ for 24 h. Expi293 cells (Gibco) were cultured following the manufacturer's instructions.

Immunofluorescence staining of hACE2-293T and hACE2-Cal-3

Cells were fixed, permeabilized, and blocked with 10% Normal Goat Serum (Invitrogen). ACE2 was stained with a primary antibody (Abcam #ab15348) followed by an Alexa Fluor 594-conjugated secondary antibody (Invitrogen #A-21442). Cells were counterstained with Hoechst 33342 (Thermo Scientific) and examined under Nikon Ti2-E Inverted Fluorescence Microscope.

Lentivirus packaging and transduction

293T cells were seeded at 80% confluence and transfected with psPAX2 (Addgene #12260), pMD2.G (Addgene #12259), and transfer plasmid pWPI-IRES-Puro-Ak-ACE2-TMPRSS2 (Addgene #154987) using polyethylenimine (PEI). Lentivirus-containing medium was harvested 72 h post-transfection, filtered through a 0.45 μ M filter, concentrated, and stored at -80°C. For transduction, 293T or Calu-3 cells were exposed to the concentrated lentivirus with 8 μ g/mL polybrene for 24 h to obtain human ACE2-overexpressing cell lines (hACE2-293T and hACE2-Cal-3 respectively).

Pseudovirus packaging, titration, and infection

Pseudoviruses were packaged in 293T cells using pCDH-EF1a-eFFly-eGFP (Addgene #104834) and spike-encoding plasmids, following a similar protocol to that of lentivirus. Post-packaging, pseudoviral particles were titrated using the Lenti-X qRT-PCR Titration Kit (Takara #631235) and used to infect target cells in the presence of 8 μ g/mL polybrene. Infectivity was assessed via a luciferase assay (Promega #E1501).

Protein production and purification

293T and Expi293 cells were transfected with HDM-CMV-sACE2(-Fc)(-LALA)-his plasmids using PEI and ExpiFectamine 293 Transfection Kit (Gibco) respectively. Culture supernatants were collected after 72 h and 5 days post-transfection respectively. The 293T supernatant was assessed for ACE2 and IgG1 levels using enzyme-linked immunosorbent assay (ELISA) kits (Abcam #ab235649; Invitrogen #BMS2092), and ACE2 activity was measured with a fluorometric assay (Abcam #ab273297). The Expi293 supernatant underwent Ni-NTA Agarose purification, followed by elution and buffer exchange to phosphate-buffered saline (PBS, pH 7.4). Protein concentration and integrity were verified using the Bradford method, ELISA, and sodium dodecyl sulfate-polyacrylamide gel electrophoresis (SDS-PAGE).

In vitro pseudovirus neutralization assay

Conditioned media containing sACE2, sACE2-Fc or sACE2-Fc-LALA proteins were diluted serially, mixed with pseudovirus (4×10^9 copies), and incubated at room temperature for 30 minutes [9]. The mixture was added to hACE2-293T cells in 96-well plates with duplicates with 8 μ g/mL polybrene. Transduction efficiency was assessed 48 h later via green fluorescent protein (GFP) imaging and/or luciferase assays.

Reporter-based *in vitro* ADCC and ADCP assays

The *in vitro* antibody-dependent cell-mediated cytotoxicity (ADCC) and antibody-dependent cellular phagocytosis (ADCP) activities of B5-D3(-LALA) were measured using Jurkat-Lucia NFAT-CD16 and Jurkat-Lucia NFAT-CD32 cells (InvivoGen) respectively according to the manufacturer's instructions. 293T cells transfected with pBOB-CAG-SARS-CoV-2-Spike-HA (Addgene #141347) acted

as target cells. Target cells were co-incubated with reporter cells and serially diluted B5-D3(-LALA) at 37°C for 1 h. Luciferase expression indicating CD16 and CD32 signaling was measured using QUANTI-Luc (InvivoGen).

ADCP of pseudovirus and confocal imaging

SARS-CoV-2 pseudovirus (8×10^8 copies) was incubated with sACE2-Fc and target cells in μ -Slide 18 Well chamber slides (ibidi). After 6 h, cells were fixed, blocked, and immunostained for human IgG-Fc (Abcam #ab98596), human immunodeficiency virus (HIV)-1 p24 (Invitrogen #PA5-81773), and lysosomal associated membrane protein 1 (LAMP1) (Abcam #ab25630). Secondary antibodies were applied (Invitrogen #A-21200 and #A-31573), and nuclei were stained with Hoechst. Confocal microscopy (Leica TCS SP8) and ImageJ software with the JACoP plugin were used to assess p24 fluorescence and its colocalization with LAMP1.

Western blot for spike cleavage detection

SARS-CoV-2 spike-HA tagged pseudovirus (4×10^9 copies) was incubated with M0/M1 macrophages or hACE2-293T cells for 6 h, with or without sACE2-Fc proteins. After incubation, cell lysates were processed through SDS-PAGE and transferred to polyvinylidene difluoride membranes. The membranes were blocked, incubated overnight with anti-HA (Merck Millipore #05-904) and anti- β -actin (Santa Cruz #sc-47778) primary antibodies, then with horseradish peroxidase (HRP)-conjugated secondary antibodies (Cell Signaling Technology #7076). Signals were detected using the Amersham ECL select kit on a Bio-Rad ChemiDoc MP system.

Quantitative PCR

Quantitative PCR (qPCR) was used to analyze lung RNA from mice. RNA was extracted from lung tissues using TRIzol (Invitrogen), reverse-transcribed (Applied Biosystems #4368813), and amplified using the TB Green Premix Ex Taq II kit (Takara), normalized to mouse *Gapdh* using the $2^{-\Delta\Delta C_t}$ method. qPCRs were run in duplicates on 384-well plates, with specific primers listed in Supplementary Table 1.

SARS-CoV-2 virus

Experiments with live SARS-CoV-2 were performed at the BSL-3 core facility (LKS Faculty of Medicine, HKU). The BetaCoV/Hong Kong/VM20001061/2020 virus, here regarded as the wild type strain of SARS-CoV-2 (Wuhan-Hu-1), was isolated from the nasopharyngeal aspirate and throat swab of a confirmed patient with COVID-19 in Hong Kong (GISAID identifier EPI_ISL_412028). The SARS-CoV-2 variants were isolated from clinical specimens in Hong Kong. Stock viruses were prepared with Vero E6 cells cultured in infection medium (DMEM supplemented with 2% FBS and 1% PS).

Median tissue culture infectious dose (TCID₅₀) assay

Vero E6 cells pre-seeded in 96-well plates were infected with serially diluted virus stocks or mouse lung homogenates in infection medium. After 72 h incubation, cytopathic effects (CPEs) were observed under a microscope to calculate titers using the Reed–Muench method.

Plaque-reduction neutralization test (PRNT) assay

Proteins were serially diluted and pre-incubated with SARS-CoV-2 variants, followed by addition to Vero E6 cells seeded in 6-well plates. After incubation, cells were overlaid with agarose, fixed with formalin, and stained with crystal violet. Plaque counts were used to calculate percentage neutralization and half maximal inhibitory concentration (IC₅₀) values.

Animal experiments

Experiments on protein-only administration in mice were carried out in the Animal Holding Core of the School of Biomedical Sciences, CUHK. Experiments involving SARS-CoV-2 infection in K18-hACE2 mice were conducted within the confines of the Biosafety Level 3 (BSL-3) core facility located at the Li Ka Shing Faculty of Medicine, HKU. Experiments were conducted according to ethical practices to minimize animal distress.

SARS-CoV-2 infection in mice

Female K18-hACE2 mice, aged 10–12 months or 2–3 months, were intranasally inoculated with 1×10^4 plaque-forming unit (PFU) of SARS-CoV-2 Wuhan-Hu-1. Treatment with B5-D3 protein was administered intranasally at 2.5 mg/kg or intravenously at 15 mg/kg, at various time points relative to the viral challenge (6 h before, 24 h before, or 24 h after). Vehicle control groups received PBS 6 h before viral challenge. Survival and weight were monitored daily for 14 days. For the older mice, lung samples were collected from one mouse from each group at 4 days post-infection (dpi) for analysis; younger mice had plasma collected for neutralizing antibody analysis at 14 dpi. Another batch of young mice also received B5-D3(-LALA) protein pre-inoculation, with lungs analyzed post-inoculation for RNA, viral load, and histopathology. A control group of non-infected mice was used to assess baseline effects of B5-D3 on lung tissue.

Neutralization assay for antibody titration

Vero E6 cells were pre-seeded on 96-well plates 24 h before infection. On the day of infection, the growth medium of the cells was changed to infection medium. The plasma samples were serially 2-fold diluted with infection medium from a starting dilution of 1:10. The plasma was then pre-incubated with 100 TCID₅₀ of SARS-CoV-2 for 1 h at room temperature before being inoculated to the seeded Vero E6 cells in quadruplicates. At 72 h after inoculation, CPEs of the cells were observed with optical microscopy. Neutralizing antibody titers against SARS-CoV-2 were expressed as the reciprocal of the highest dilution of plasma showing no CPEs in all 4 wells. Uninfected cell monolayers were used as toxicity control.

Histology

Mouse tissues were fixed in 10% formalin, embedded in paraffin, and sectioned at 5 μ m. Sections of different organs were deparaffinized and underwent hematoxylin and eosin (H&E) staining. For immunohistochemistry (IHC) staining, lung sections underwent antigen retrieval, endogenous peroxidase blocking, and were incubated with primary antibodies against the SARS-CoV/SARS-CoV-2 nucleocapsid protein (Sino Biological #40143-T62) overnight. After washing, sections were stained with the anti-rabbit VECTASTAIN Elite ABC-HRP Kit (Vector Laboratories), developed with 3,3'-Diaminobenzidine (Sigma #D4293), and counterstained with Mayer's hematoxylin.

RNA-Seq and data analysis

Total RNA was extracted from lung tissues using TRIzol and processed into transcriptome libraries with the TruSeq RNA Library Prep Kit (illumina). Sequencing was performed on the NovaSeq 6000 sequencer (illumina) using a 150-base pair paired-end configuration. Sequencing data were processed with fastp for quality control [6], then aligned to both the mouse (Ensembl GRCm39) and SARS-CoV-2 (NCBI NC_045512v2) genomes using STAR [29]. Pearson correlation and groupwise comparisons were conducted in R: gene expression was quantified and analyzed for differential expression using DESeq2 [27]; up/downregulated gene enrichment and Gene Set Enrichment Analysis (GSEA) [39] was performed using the clusterProfiler package [48].

Tracking B5-D3 bio-distribution in mice

B5-D3 were conjugated with Alexa Fluor 750 dye (B5-D3-AF750) as described [21]. In brief, 2 mg/mL solution of B5-D3 protein in 0.1M NaHCO₃ was reacted with Alexa Fluor 750 succinimidyl ester (Thermo Fisher Scientific) at room temperature for 1 h. Unreacted dye was removed by dialysis in PBS. All procedures were performed under dimmed light. Female K18-hACE2 mice, aged 2-3 month, were administered intranasally with B5-D3-AF750 (2.5 mg/kg). The mice were imaged at predetermined time points after administration (fluorescence ex = 745 nm, em = 800 nm, auto-exposure setting) using an IVIS Spectrum CT Imager (Perkin Elmer). At the time of euthanasia, 50 µl of urine and blood, the brain, nasal cavity, trachea, lung, heart, liver, spleen, kidney, and urinary bladder samples were excised and imaged. Regions of interest (ROIs) were drawn, and average radiance (p/s/cm²/sr) was measured. All images were processed using Living Image software (Perkin Elmer) and the same fluorescence threshold was applied for group comparison.

Flow cytometry analysis and confocal microscopic imaging of BALF cells

Mice were sacrificed via anesthetics overdose. Bronchoalveolar lavage was performed by intratracheally rinsing the lungs with 1 mL of ice-cold Hanks' Balanced Salt Solution (HBSS, Gibco) containing 100 µM ethylenediaminetetraacetic acid for four repeats. Bronchoalveolar lavage fluid (BALF) was then centrifuged and treated with ammonium-chloride-potassium red blood cell lysing buffer. Cell pellets were washed with PBS and stained with the Fixable Viability Stain 440UV dye (BD #566332). Next, the cells were blocked with CD16/CD32 monoclonal antibody (Invitrogen #14-0161-85) and stained with antibodies targeting the following molecules: CD45 (BD #568336), Siglec-F (BD #564514), CD11b (BD #612800), CD11c (BD #751265), Ly6G (BD #563005), I-A/I-E major histocompatibility complex class II (MHC-II) (BD #750171), F4/80 (BD #570288), Ly6C (BD #755198), and CD3 (BD #555275). Stained BALF cells were analyzed using the BD FACSymphony A5.2 SORP Flow Cell Analyzer, and the results were analyzed using FlowJo v10.10.

For microscopic inspections, BALF cells were seeded in poly-d-lysine-coated chamber slides and stained with antibodies targeting Siglec-F (BD #564514) and human IgG-Fc (Abcam #ab98596). Secondary antibody (Invitrogen #A-11006) was applied to amplify Siglec-F signal. Stained BALF cells were then counterstained with Hoechst and examined by confocal microscopy (Leica TCS SP8).

Statistical Analysis

Assays including *in vitro* neutralization, PRNT, ADCC, and ADCP were conducted in technical duplicates. Results were analyzed in GraphPad Prism version 9 using nonlinear regression to calculate IC₅₀ or half maximal effective concentration (EC₅₀) values. Transcriptomic analyses were performed using R, with details provided in figure captions. All other statistical analyses utilized GraphPad Prism version 9 with a significance threshold set at *p* value (*p*) < 0.05.

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Additional information

Availability of data and materials

All data associated with this study are available in the main text or the supplementary materials. The RNA-seq data generated in this study have been deposited in the NCBI Sequence Read Archive database under accession code PRJNA1054508. Constructs of diverse sACE2-Fc mutants and SARS-CoV-2 spikes are available upon request after completion and approval of a material transfer agreement by contacting fengbo@cuhk.edu.hk.

Ethics approval and consent to participate

All animal procedures were ethically approved by The Chinese University of Hong Kong (CUHK)'s Animal Experimentation Ethics Committee (approval number: 20-226-MIS) and The University of Hong Kong (HKU)'s Committee on the Use of Live Animals in Teaching and Research (approval number: 5511-20).

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Authors' contributions

Jingyi W. and J.L. constructed the sACE2-Fc mutants and performed characterization analysis; A.W.C. performed the PRNTs and data analysis; Jingyi W., A.W.C., and J.L. performed the mouse infection experiments and data analysis; B.L., J.Q. and J.R. produced recombinant sACE2-Fc proteins; Jingyi W. and Junkang W. performed RNA-Seq analysis; Jingyi W. and J.Q. performed protein labeling, *in vivo* tracing of labeled protein, and flow cytometry analysis of BALF cells; J.L. performed THP-1 and Calu-3 experiments and confocal microscopic analysis. Junkang W. and J.L. performed protein structure prediction and visualization. Jingyi W., J.L., L.L.P. and B.F. conceived the project, designed experiments, and wrote the manuscript. Y.X., T.B., L.L.P. and B.F. revised the manuscript. All authors read and approved the final manuscript.

Additional files

Supplementary information. Figures S1 to S15 and Table S1. [↗](#)

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Reviewer #1 (Public review):

Summary:

This manuscript by Wang et al. describes the development of an optimized soluble ACE2-Fc fusion protein, B5-D3, for intranasal prophylaxis against SARS-CoV-2. As shown, B5-D3 conferred protection not only by acting as a neutralizing decoy, but also by redirecting virus-decoy complexes to phagocytic cells for lysosomal degradation. The authors showed complete in vivo protection in K18-hACE2 mice and investigated the underlying mechanism by a combination of Fc-mutant controls, transcriptomics, biodistribution studies, and in vitro assays.

Strengths:

The major strength of this work is the identification of a novel antiviral approach with broad-spectrum and beyond simple neutralization. Mutant ACE2 enables broad and potent binding

activity with the S proteins of SARS-CoV-2 variants, while the fused Fc part mediates phagocytosis to clear the viral particles. The conceptual advance of this ACE2-Fc combination is convincingly validated by in vivo protection data and by the completely abrogated protection of Fc LALA mutant.

Weaknesses:

Some aspects could be further modified.

(1) A previously reported ACE2 decamer (DOI: 10.1080/22221751.2023.2275598) needs to be mentioned and compared in the Discussion part.

(2) Limitations of this study, such as off-target binding and potential immunogenicity, should also be discussed.

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Reviewer #2 (Public review):

Summary:

Wang et al. engineered an optimized ACE2 mutant by introducing two mutations (T92Q and H374N) and fused this ACE2 mutant to human IgG1-Fc (B5-D3). Experimental results suggest that B5-D3 exhibits broad-spectrum neutralization capacity and confers effective protection upon intranasal administration in SARS-CoV-2-infected K18-hACE2 mice. Transcriptomic analysis suggests that B5-D3 induces early immune activation in lung tissues of infected mice. Fluorescence-based bio-distribution assay further indicates rapid accumulation of B5-D3 in the respiratory tract, particularly in airway macrophages. Further investigation shows that B5-D3 promotes viral phagocytic clearance by macrophages via an Fc-mediated effector function, namely antibody-dependent cellular phagocytosis (ADCP), while simultaneously blocking ACE2-mediated viral infection in epithelial cells. These results provide insights into improving decoy treatments against SARS-CoV-2 and other potential respiratory viruses.

Strengths:

The protective effect of this ACE2-Fc fusion protein against SARS-CoV-2 infection has been evaluated in a quite comprehensive way.

Weaknesses:

(1) The paper lacks an explanation regarding the reason for the combination of mutations listed in Supplementary Figure 2b. For example, for the mutations that enhance spike protein binding, B2-B6 does not fully align with the mutations listed in Table S1 of Reference 4, yet no specific criteria are provided. Second, for the mutations that abolished enzymatic activity, while D1 and D2, D3, D4, and D5 are cited from References 12, 11, and 33, respectively, the reason for combining D3 and D4 into A2, and D1 and D2 into A3 remains unexplained. It is also unclear whether some of these other possible combinations have been tested. Furthermore, for the B5-derived mutations, only double-mutant combinations with D1-D5 are tested, with no attempt made to evaluate triple mutations involving A2 or A3.

(2) Figures 1b, 1d, and 1e lack statistical analyses, making it difficult to determine whether B5 and D3 exhibit significant advantages. For Wuhan-Hu-1 strain, B2 and B5 are similar, and for D614G strain, B2, B3, B4, B5, and B6 display comparable results. However, only the glycosylation-related single mutant B5 is chosen for further combinatorial constructs. Moreover, for VOC/VOI strains, B5 is superior to B5-D3; for the Alpha strain, B5-D4 and B5-D5 are superior to B5-D3; and for the Delta and Lambda strains, B5-D5 is superior to B5-D3.

These observations further highlight the need for a clearer explanation of the selection strategy.

(3) Figure 1e does not specify the construct form of the control hIgG1, namely whether it is an hIgG1 Fc fragment or a full-length hIgG1 protein. If the full-length form is used, the design of its Fab region should be clarified to ensure the accuracy and comparability of the experimental control.

(4) In Figure 2a, all three PBS control mice died, whereas in Figure 2f, three out of five PBS control mice died, with the remaining showing gradual weight recovery. This discrepancy may reflect individual immune variations within the control groups, and it is necessary to clarify whether potential autoimmune factors could have affected the comparability of the results. Also, the mouse experiments suffer from insufficient sample sizes, which affects the statistical power and reliability of the results. In Figure 2a, each group contains only 4 replicates, one of which was used for lung tissue sampling. As a result, body weight monitoring data is derived from only 3 mice per group (the figure legend indicating $n=4$ should be corrected to $n=3$). Such a small sample size limits the robustness of the conclusions. Similarly, in Figure 2f, although each group has 5 replicates, body weight data are presented for only 4 mice, with no explanation provided for the exclusion of the fifth mouse. Furthermore, the lung tissue experiments in Figure 3a include only 3 replicates, which is also inadequate.

(5) Compared to 6 hours, intranasal administration of B5-D3 at 24 hours before viral infection results in reduced protective efficacy. However, only survival and body weight data are provided, with no supporting evidence from virological assays such as viral titer measurement. Therefore, the long-term effectiveness lacks sufficient experimental validation.

(6) In Figures 3b and 3c, viral spike (S) and nucleocapsid (N) RNA relative expression levels are quantified by qPCR. The results show significant individual variation within the B5-D3-LALA treatment group: one mouse exhibits high S and N expression, while the other two show low expression. Viral load levels are also inconsistent: two mice have high viral loads, and one has a low viral load. Due to this variability, the available data are insufficient to robustly support the conclusion.

(7) Figure 3e: "H&E staining indicated alveolar thickening in all groups," including the Mock group. Since the Mock group did not receive virus or active drug treatment, this observed change may result from local tissue reaction induced by the intranasal inoculation procedure itself, rather than specific immune activation. A control group (no manipulation) should be set to rule out potential confounding effects of the experimental procedure on tissue morphology, thereby allowing a more accurate assessment of the drug's effects.

(8) In Supplementary Figure 11b, a considerable number of alveolar macrophages (AMs) are observed in both the PBS and B5-D3 groups. This makes it difficult to determine whether the observed accumulation is specifically induced by B5-D3.

(9) In the flow cytometry experiment shown in Figure 5, the PBS control group is not labeled with AF750, which necessarily results in a value of zero for "B5-D3+ cells" on the y-axis. An appropriate control (e.g., hIgG1-Fc labeled with AF750) should be included.

(10) The Methods section: a more detailed description of the experimental procedures involving HIV p24 and SARS-CoV-2 should be included.

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Reviewer #3 (Public review):**Strengths:**

The core strength of this study lies in its innovative demonstration that an engineered sACE2-Fc fusion redirects virus-decoy complexes to Fc-mediated phagocytosis and lysosomal clearance in macrophages, revealing a distinct antiviral mechanism beyond traditional neutralization. Its complete prophylactic protection in animal models and precise targeting of airway phagocytes establish a novel therapeutic paradigm against SARS-CoV-2 variants and future respiratory viruses.

Weaknesses:

The study attributes the complete antiviral protection to Fc-mediated phagocytic clearance, a central claim that requires more rigorous experimental validation. The observation that abrogating Fc functions compromises protection could be confounded by potential alterations in the protein's stability, half-life, or overall structure. To firmly establish this mechanism, it is crucial to include a control molecule with a mutated Fc region that lacks FcγR binding while preserving the Fc structure itself. Without this critical control, the conclusion that phagocytic clearance is the primary mechanism remains inadequately supported. The strategy of deliberately targeting virus-decoy complexes to phagocytes via Fc receptors inherently raises the question of Antibody-Dependent Enhancement (ADE) of disease. While the authors demonstrate a lack of productive infection in macrophages, this only addresses one facet of ADE. The risk of Fc-mediated exacerbation of inflammation (ADE) remains a critical concern. The manuscript would be significantly strengthened by a direct discussion of this risk and by including data, such as cytokine profiling from treated macrophages, to more comprehensively address the safety profile of this approach. The exclusive use of the K18-hACE2 mouse model, which exhibits severe disease, limits the generalizability of the findings. The "complete protection" observed may not translate to models with more robust and naturalistic immune responses or to human physiology. Furthermore, the lack of data on circulating SARS-CoV-2 variants is a concern. The concept of sACE2-Fc fusion proteins as decoy receptors is not novel, and numerous similar constructs have been previously reported. The manuscript would benefit from a clearer demonstration of how the optimized B5-D3 mutant represents a significant advance over existing sACE2-Fc designs. A direct comparative analysis with previously published benchmarks, particularly in terms of neutralizing potency, Fc effector function strength, and in vivo efficacy, is necessary to establish the incremental value and novelty of this specific agent.

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